

AKHIL MM *FULL STACK DEVELOPER*

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🔗 Portfolio

ABOUT ME

Full-stack developer with hands-on experience building real-world full-stack applications and backend systems. Developed projects including a ride-booking platform, secure chat application with AI features, e-commerce platform, and accessibility-focused solutions. Interested in scalable system design, real-time applications, and solving practical problems through software. Strong focus on writing maintainable code, continuous learning, and building products with real-world impact.

Technologies

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Redux, Next.js, Tailwind CSS, Material UI, HTML, CSS

Backend: Node.js, Express.js, RESTful APIs, Socket.IO [↗](#)

Database & Caching: MongoDB, Redis (Upstash Redis), MongoDB Atlas Vector Search

Cloud & Deployment: AWS EC2, AWS S3, Google Cloud Platform (GCP), NGINX, PM2, Vercel

Tools & Technologies: Git, GitHub, Postman, Docker, Cloudinary, Stripe, Razorpay

Projects

Sarathi – Driver-On-Demand Platform [↗](#)

Built a full-stack driver-on-demand platform with booking management, secure payments, and real-time chat/video communication features.

Technologies: TypeScript, Node.js, Express.js, MongoDB, React.js, Tailwind CSS, Material UI, Socket.IO [↗](#), WebRTC, Redis, Stripe, Cloudinary, AWS EC2

- Architected the backend using TypeScript, Express.js, MongoDB, and Tsyringe dependency injection following Clean Architecture and SOLID principles.
- Implemented JWT authentication, OTP email verification via Nodemailer, and Redis-based temporary registration storage for user and driver onboarding.
- Developed real-time chat and video calling features using Socket.IO [↗](#) and raw WebRTC supporting text, emojis, images, PDFs, and file sharing.
- Integrated Stripe Connected Accounts for driver onboarding and automated payment split handling between platform and drivers.
- Implemented secure media uploads using Cloudinary signed uploads with server-generated signatures.
- Integrated Google Maps API for location autocomplete and road-distance calculation between pickup and destination points.
- Built responsive frontend interfaces using React.js, TypeScript, Tailwind CSS, and Material UI.
- Deployed backend on AWS EC2 using PM2 and NGINX, and hosted frontend on Vercel.

AI-Powered Real-Time Group Chat Platform [↗](#)

Built a scalable real-time chat platform with Gemini AI integration using Retrieval-Augmented Generation (RAG), semantic vector search, and high-performance rendering for large chat histories.

Technologies: TypeScript, Node.js, Express.js, React.js, MongoDB Atlas Vector Search, Socket.IO [↗](#), Google Gemini API, Tailwind CSS, AWS S3, GCP

- Integrated Google Gemini API with a custom RAG pipeline using MongoDB Atlas Vector Search and gemini-embedding-2 for context-aware AI responses.

- Architected the backend using TypeScript and Clean Architecture principles with modular Use Cases, Repositories, Services, and Controllers.
- Implemented high-performance virtualized message rendering using TanStack Virtual with optimized infinite scrolling, minimized re-renders, and efficient handling of large chat histories.
- Developed scalable real-time messaging features using Socket.IO [↗](#) including AI-assisted conversations, read receipts, intelligent message grouping, media sharing, and real-time event handling.
- Implemented JWT access/refresh token rotation, Google OAuth 2.0 authentication, and AWS S3 presigned URLs for secure direct media uploads.
- Built responsive and animated frontend interfaces using React.js, Tailwind CSS, and Framer Motion.
- Deployed backend services on Google Cloud Platform (GCP) and hosted the frontend on Vercel.

EYEGAZE – Full-Stack E-Commerce Platform [↗](#)

Built a full-stack eyewear e-commerce platform with product management, secure payments, analytics dashboard, and inventory management features.

Technologies: Node.js, Express.js, MongoDB, EJS, Bootstrap, Chart.js, Razorpay

- Developed secure authentication and authorization workflows using session-based authentication and custom middleware.
- Integrated Razorpay payment gateway for secure checkout and order processing.
- Built an admin dashboard using Chart.js to visualize revenue, user growth, and sales analytics.
- Implemented product, category, inventory, offer, and coupon management modules for streamlined store administration.
- Developed optimized product image upload handling using Multer for multi-image processing and catalog management.
- Designed responsive frontend interfaces using EJS, Bootstrap, HTML, and CSS.

Education

Bachelor of Science (B.Sc [↗](#) .) in Mathematics
with Complementary Studies in Computer Science
University of Calicut | 2019 – 2022